

1 Solution — GIAM 3.2

Problem 7

1.1 Problem

Show that

$$\binom{n}{k} \binom{k}{r} = \binom{n}{r} \binom{n-r}{k-r}$$

for all integers r, k, n with $r \leq k \leq n$.

1.2 Proof

Proof. Expand both sides using $\binom{m}{j} = \frac{m!}{j!(m-j)!}$.

Left side.

$$\binom{n}{k} \binom{k}{r} = \frac{n!}{k!(n-k)!} \cdot \frac{k!}{r!(k-r)!} = \frac{n!}{r!(k-r)!(n-k)!},$$

the $k!$ cancelling.

Right side. Note first that $(n-r) - (k-r) = n-k$, so

$$\binom{n}{r} \binom{n-r}{k-r} = \frac{n!}{r!(n-r)!} \cdot \frac{(n-r)!}{(k-r)!(n-k)!} = \frac{n!}{r!(k-r)!(n-k)!},$$

the $(n-r)!$ cancelling.

The two expressions are identical, so the sides are equal. ■

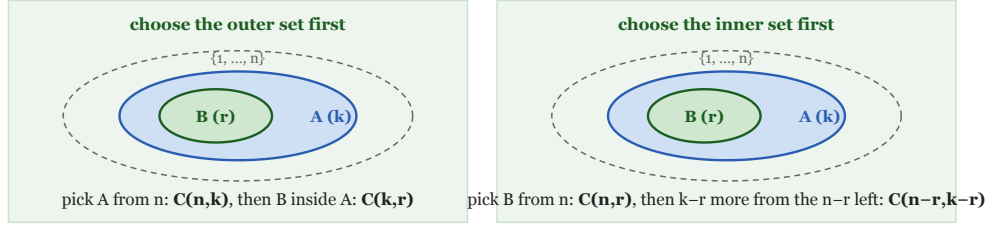
The condition $r \leq k \leq n$ guarantees r , $k-r$ and $n-k$ are all non-negative, so every factorial above is defined. Verified for every valid triple with $n \leq 25$.

1.3 What the Identity Says

Both sides collapse to the same trinomial coefficient — and both count the same nested pairs.

Expand each side into factorials	
left side $n! / (k!(n-k)! \cdot k! / (r!(k-r)!))$ <i>the $k!$ cancels</i> $= n! / (r! (k-r)! (n-k)!)$	right side $n! / (r!(n-r)! \cdot (n-r)! / ((k-r)!(n-k)!))$ <i>the $(n-r)!$ cancels, and $(n-r)-(k-r) = n-k$</i> $= n! / (r! (k-r)! (n-k)!)$
identical ■	

Why it is true: both sides count nested pairs $B \subset A \subset \{1, \dots, n\}$ with $|A| = k$, $|B| = r$



same collection counted two ways, so the two products are equal

Equivalently: a nested pair just splits $\{1, \dots, n\}$ into three labelled blocks

B (size r) · A minus B (size $k-r$) · everything outside A (size $n-k$)
and the number of such splits is the trinomial coefficient $n! / (r! (k-r)! (n-k)!)$

Worked check, $n = 7, k = 4, r = 2$: $C(7, 4) \cdot C(4, 2) = 35 \cdot 6 = 210$ and $C(7, 2) \cdot C(5, 2) = 21 \cdot 10 = 210$

Verified for every valid triple with $n \leq 25$ — no mismatches.

Figure 1.1: Top panel, expanding each side into factorials. Left side: n factorial over k factorial times $(n \text{ minus } k)$ factorial, times k factorial over r factorial times $(k \text{ minus } r)$ factorial; the k factorial cancels, leaving n factorial over r factorial times $(k \text{ minus } r)$ factorial times $(n \text{ minus } k)$ factorial. Right side: n factorial over r factorial times $(n \text{ minus } r)$ factorial, times $(n \text{ minus } r)$ factorial over $(k \text{ minus } r)$ factorial times $(n \text{ minus } k)$ factorial; the $(n \text{ minus } r)$ factorial cancels, and $(n \text{ minus } r) \text{ minus } (k \text{ minus } r)$ equals $n \text{ minus } k$, leaving the same expression. Identical, so the proof is done. Below, the combinatorial reason: both sides count nested pairs B inside A inside the set from 1 to n , with A of size k and B of size r . Left diagram: choose the outer set first, showing a dashed universe containing a blue ellipse A containing a green ellipse B — pick A from n in $C(n, k)$ ways, then B inside A in $C(k, r)$ ways. Right diagram: choose the inner set first, the same nested picture — pick B from n in $C(n, r)$ ways, then $k \text{ minus } r$ more elements from the $n \text{ minus } r$ remaining, in $C(n \text{ minus } r, k \text{ minus } r)$ ways. A green banner: the same collection counted two ways, so the two products are equal. A final panel: equivalently a nested pair just splits the set into three labelled blocks, B of size r , $A \text{ minus } B$ of size $k \text{ minus } r$, and everything outside A of size $n \text{ minus } k$, and the number of such splits is the trinomial coefficient. Worked check for

n equals 7, k equals 4, r equals 2: both sides give 210. Verified for every valid triple with n at most 25.

The algebra is short, but it hides the reason. Both sides count the **same collection**: pairs of nested sets

$$B \subseteq A \subseteq \{1, 2, \dots, n\}, \quad |A| = k, \quad |B| = r.$$

- **Left side.** Choose A first: $\binom{n}{k}$ ways. Then choose B from inside A : $\binom{k}{r}$ ways.
- **Right side.** Choose B first: $\binom{n}{r}$ ways. Then enlarge it to A by picking the remaining $k - r$ elements from the $n - r$ not already in B : $\binom{n-r}{k-r}$ ways.

Two ways of counting one set must agree. A brute-force count for $n = 6$, $k = 4$, $r = 2$ gives 90 nested pairs, matching $\binom{6}{4} \binom{4}{2} = \binom{6}{2} \binom{4}{2} = 90$.

1.4 Remark — Both Sides Are a Trinomial Coefficient

The common value is worth naming:

$$\binom{n}{k} \binom{k}{r} = \frac{n!}{r! (k-r)! (n-k)!} = \binom{n}{r, k-r, n-k},$$

a **multinomial coefficient**. It counts the ways to split $\{1, \dots, n\}$ into three labelled blocks of sizes r , $k - r$ and $n - k$ — which is exactly what a nested pair $B \subseteq A$ is:

block	what it is	size
first	B	r
second	$A \setminus B$	$k - r$
third	outside A	$n - k$

Table 1.1

Seen this way the identity is not two facts but one object described from two angles. The symmetric form also makes clear why the three quantities r , $k - r$, $n - k$ must be non-negative — hence the hypothesis $r \leq k \leq n$.

1.5 Remark — Which Proof Is Better?

Both are complete, and they answer different questions.

The **factorial proof** is the one to write when you need certainty: it is mechanical, checkable line by line, and needs no cleverness. It verifies *that* the identity holds.

The **counting proof** explains *why*. It also generalises without effort — the same “choose in either order” argument immediately gives the m -fold version

$$\binom{n}{k_1} \binom{k_1}{k_2} \cdots \binom{k_{m-1}}{k_m} = \frac{n!}{k_m! (k_{m-1} - k_m)! \cdots (n - k_1)!}$$

for a chain of nested sets, whereas pushing the factorial computation that far is tedious. This is the recurring trade-off: algebra confirms, combinatorics illuminates.

1.6 Remark — The Step $(n - r) - (k - r) = n - k$

The one place the right-hand computation could go wrong is the second factorial. Applying $\binom{m}{j} = \frac{m!}{j!(m-j)!}$ with $m = n - r$ and $j = k - r$ gives a bottom factor $(m - j)! = ((n - r) - (k - r))!$, and it takes a moment to see that this simplifies to $(n - k)!$.

That simplification is what makes the two sides match — and it is also exactly the combinatorial statement that after removing B (size r) and then the rest of A (size $k - r$), what remains outside A has size $n - k$. The algebra and the picture are checking the same bookkeeping.

1.7 Remark — Special Cases Worth Recognising

Setting the parameters to extremes recovers familiar facts:

choice	identity becomes
$r = 0$	$\binom{n}{k} \cdot 1 = 1 \cdot \binom{n}{k}$ — trivial

$r = k$	$\binom{n}{k} \cdot 1 = \binom{n}{k} \binom{n-k}{0} \text{ — trivial}$
$k = n$	$1 \cdot \binom{n}{r} = \binom{n}{r} \cdot 1 \text{ — trivial}$
$r = 1$	$k \binom{n}{k} = n \binom{n-1}{k-1}$

Table 1.2

The last is the genuinely useful one — the **committee-and-chair identity**. Reading it combinatorially: choosing a k -member committee and then its chair ($k \binom{n}{k}$) is the same as choosing the chair first and then $k - 1$ others from the remaining $n - 1$ ($n \binom{n-1}{k-1}$). It is the standard tool for evaluating sums like $\sum_k k \binom{n}{k}$.

1.8 Remark — The Hypothesis $r \leq k \leq n$

Two inequalities, each doing a job. $r \leq k$ keeps $\binom{k}{r}$ and $(k - r)!$ meaningful; $k \leq n$ does the same for $\binom{n}{k}$ and $(n - k)!$. Together they make all three block sizes non-negative.

With the common convention $\binom{m}{j} = 0$ whenever $j < 0$ or $j > m$, the identity in fact survives outside this range — both sides simply become 0. But that convention has to be *adopted*; stating the hypothesis is the honest alternative, and it matches the pattern seen throughout this chapter, where the hypothesis exists precisely to keep every expression in the statement well defined.